The developments discussed in this and the previous sections represent the first, somewhat tentative steps toward understanding the cosmological implications of string/M-theory. During the coming years, as the nonperturbative tools of string/M-theory are sharpened, physicists anticipate that some of the most profound insights will emerge from their application to cosmological questions.

But without currently having methods that are sufficiently powerful to understand cosmology according to string theory fully, it is worthwhile to think about some general considerations concerning the possible role of cosmology in the search for the ultimate theory. We caution that some of these ideas are of a more speculative nature than much of what we have discussed previously, but they do raise issues that any purported final theory may one day have to address.

Cosmological Speculation and the Ultimate Theory

Cosmology has the ability to grab hold of us at a deep, visceral level because an understanding of how things began feels—at least to some—like the closest we may ever come to understanding why they began. That is not to say that modern science provides a connection between the question of how and the question of why—it doesn't—and it may well be that no such scientific connection is ever found. But the study of cosmology does hold the promise of giving us our most complete understanding of the arena of the why—the birth of the universe—and this at least allows for a scientifically informed view of the frame within which the questions are asked. Sometimes attaining the deepest familiarity with a question is our best substitute for actually having the answer.

In the context of searching for the ultimate theory, these lofty reflections on cosmology give way to far more concrete considerations. The way things in the universe appear to us today—way on the far right-hand side of the time line in Figure 14.1—depends upon the fundamental laws of physics, to be sure, but it may also depend on aspects of cosmological evolution, from the far left-hand side of the time line, that potentially lie outside the scope of even the deepest theory.

It's not hard to imagine how this might be. Think of what happens, for example, when you toss a ball in the air. The laws of gravity govern the ball's subsequent motion, but we can't predict where the